

Yuehang Lin

HCI/VR/Haptics Researcher | Interdisciplinary Research Engineer

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A highly effective Interdisciplinary (Maths, English, Computer Science) Research Engineer specializing in VR, Haptics, and Human-Computer Interaction (HCI). I possess a holistic, data-driven skill set spanning the entire research pipeline: from high-fidelity simulation (Unity, C#) and hardware prototyping (Fusion 360, sensors) to rigorous experimental evaluation. My core expertise lies in translating complex research problems into reliable, scalable, and cross-disciplinary interactive systems, and I am recognized as a fast learner who adapts quickly to new tools and domains.

EDUCATION

(MSc) Advanced Computer Science University of Birmingham (QS World Ranking: 76)	Sept 2024 - Dec 2025
(BSc) Information & Computing Science Taiyuan University of Technology (211 Project)	Sept 2017 - June 2021
(BA) English (Dual Degree) Taiyuan University of Technology (211 Project)	Sept 2019 - June 2021

RESEARCH EXPERIENCE

Research Associate:

University of Birmingham – VR Lab

October 2025 – Present, Birmingham, UK

- Contributed to immersive projection environments for research meetings, integrating real-time visualization with multi-sensor spatial setups.
- Extended a Unity-based virtual grasp simulation into a real robotic gripper system, using mechanically weighted off-center objects to study haptic feedback and grasp stability in realistic environments.
- Implemented interaction pipelines linking Unity simulation, haptic devices, and robotic control systems for real-time experimental testing.

Research Project:

Haptic Slip Rendering for Improved Grasp Stability in VR Teleoperation (MSc Thesis) Supervised by Prof. Eyal Ofek

- Pioneered the Haptic Grasp VR teleoperation system, delivering localized, directional haptic slip feedback to address a critical gap in stable robotic manipulation cues.
- Engineered an end-to-end real-time VR-haptics closed-loop pipeline (Unity/C#, custom hardware) and conducted a rigorous controlled study (N=21), analyzing over 2 million frames of interaction data.
- Critically demonstrated performance gains: Participants achieved their first successful grasp ~30% faster ($p=0.001$) with haptic cues. The Haptic condition alone reduced angular error from a $\sim 45^\circ$ plateau to $\sim 18^\circ$ by the 5th attempt.
- Validated significant subjective benefits: Multimodal feedback enhanced user confidence and clarity, achieving mean subjective ratings >6 (vs. ~ 4.5 unimodal conditions), confirming the system's perceptual value.

Research Assistant:

University of Birmingham – VR Lab

Jun 2025 - Sep 2025, Birmingham, UK

- Served as a core contributor to the RemoteHapticGrasp project, focusing on enhancing robotic teleoperation through directional slip cues with the utmost scientific rigor (as confirmed by the certificate).
- Designed the system architecture and implemented real-time interaction code, successfully engineering the hardware-software link between the Unity VR simulation and external haptic devices.
- Co-designed user study protocols and data collection modules, collecting interaction frames data to quantitatively assess grasp stability and the perceptual benefits of the novel feedback.

Research Assistant:

OBI Robotics Limited | Research Assistant

Dec 2024 - May 2025, Birmingham, UK

- Accelerated haptic research cycles by rapidly mastering Unity and Fusion 360 to develop high-fidelity simulation environments and sensor-based prototypes, reducing prototype iteration time by $\sim 40\%$.
- Led hardware engineering for the “Crystal MaNia” interactive art installation, bridging abstract artistic concepts with robust HCI design. Specifically, modeled and integrated multi-axis sensor prototypes to control sound processes through crystal manipulation.

- Contributed to remote robotic grasping and haptic rendering projects by providing cross-disciplinary support, implementing reliable software modules (C#/Python) and robust hardware interfaces essential for long-term experimental validation.

PUBLICATIONS, THESIS, AND MANUSCRIPTS IN PREPARATION

Master's Thesis

Haptic Slip Rendering for Improved Grasp Stability in VR Teleoperation

(Supervisor: **Eyal Ofek**, University of Birmingham, 2025)

Manuscripts in Preparation / Working Papers

Remote HapticGrasp: Remote Robot Grip Guidance using Haptic Rendering

Yuehang Lin, Daniele Giunchi, Diar Abdlkarim, Massimiliano Di Luca, Eyal Ofek.

Status: Working Paper. Initial version intended for submission to a top-tier HCI conference (e.g., CHI/UIST). Research is currently being extended to robotic hardware integration for future publication.

TECHNICAL SKILLS

I. Immersive Prototyping & Software: Unity (C#), Virtual Reality (VR), AR/VR SDKs (e.g., OpenXR), Real-Time Physics Simulation, Visual Studio, Git/GitHub/GitLab, Version Control.

II. Haptics & Hardware Engineering: Arduino IDE/Microcontrollers (e.g., Nano 33 BLE), Sensor & Actuator Integration, Fusion 360 (3D Modeling), 3D Printing (FDM), Circuit Soldering, Hardware-Software Integration, Serial/BLE Communication.

III. Research & Data Analysis: User Study Design (Controlled Experimentation), Quantitative Data Analysis, Statistical Testing, Colab, Python (Pandas, NumPy, Matplotlib), Edge Impulse (ML Model Deployment), LaTeX/Overleaf.

IV. Core Programming & Web: C#, Python, Java, HTML, Flutter (Mobile/IoT UI).

PROJECTS

Crystal Music Control Installation | Interactive Art Project

Jan 2025 - Present

- Designed sensor-based interactive modules (joystick, rotation, pressure) for artistic installations.
- Collaborated with artist Cristiana Palandri to translate abstract artistic concepts into functional interactive prototypes.
- Ensured stable performance in public exhibitions with modular, reusable design.

Fall8 | IoT-based Fall Detection System

Feb 2025 - May 2025

- Developed a prototype IoT Fall Detection System using Arduino 33 Nano with BLE and a Flutter mobile app to provide immediate alerts for elderly patients.
- Employed the Edge Impulse platform to collect sensor data and deploy a machine learning model for robust fall detection, achieving an accuracy of 95%.

VitaFit | Health Management App Concept

Oct 2024 - Dec 2024

- Designed modules for health data collection, goal setting, and personalized recommendations.
- Delivered UI prototypes focused on simplicity and usability, supporting future AI-driven features.

Interactive Learning Platform for Vocational Chinese

Jan 2024 - Aug 2024

- Built platform architecture and interactive modules for content browsing, learning, and file management.
- Implemented segmented learning mechanism, boosting course completion rate by 50%.
- Developed progress tracking and reminders, increasing task completion rate by 40%.

LEADERSHIP & CAMPUS EXPERIENCE

President, TYUT Dream Wings Roller Skating Association

Sept 2018 - Sept 2019

- Led 300+ members, organized skating festivals and large-scale events, improving community influence and cohesion.

President, Student Union, Xuzhou No.1 High School.

Sept 2015 - Sept 2016

Head of Discipline Department, Student Union, Xuzhou Xiexiu Junior High School.

Sept 2012 - Sept 2013